



1. Understand the Condition for Stability

A system is asymptotically stable if and only if the real parts of **all** eigenvalues of the closed-loop matrix (A_{cl}) are strictly negative.

The formula for the closed-loop matrix with state feedback ($u = -Kx$) is:

$$A_{cl} = A - BK$$

2. Calculate the Matrix BK

First, we multiply the Input Matrix B by the Gain Matrix K .

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad K = \begin{bmatrix} k & 0 & 0 \\ 0 & k & 0 \end{bmatrix}$$

$$BK = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} k & 0 & 0 \\ 0 & k & 0 \end{bmatrix} = \begin{bmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

3. Calculate the Closed-Loop Matrix A_{cl}

Now, we subtract BK from the system matrix A .

$$A = \begin{bmatrix} 3 & -1 & 0 \\ 0 & -4 & 2 \\ 0 & 0 & -1 \end{bmatrix}$$

$$A_{cl} = A - BK = \begin{bmatrix} 3 & -1 & 0 \\ 0 & -4 & 2 \\ 0 & 0 & -1 \end{bmatrix} - \begin{bmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$A_{cl} = \begin{bmatrix} 3-k & -1 & 0 \\ 0 & -4-k & 2 \\ 0 & 0 & -1 \end{bmatrix}$$

4. Find the Eigenvalues

To check stability, we need the eigenvalues of A_{cl} .

Notice that the matrix A_{cl} is **Upper Triangular** (all entries below the main diagonal are zero).

For a triangular matrix, the eigenvalues are simply the diagonal elements:

1. $\lambda_1 = 3 - k$
2. $\lambda_2 = -4 - k$
3. $\lambda_3 = -1$

5. Solve for Stability

For the system to be stable, all eigenvalues must be less than zero ($\lambda < 0$).

- **Check λ_3 :**

$$-1 < 0$$

(This is always true, so this part of the system is naturally stable).

- **Check λ_2 :**

$$-4 - k < 0$$

$$-4 < k$$

$$k > -4$$

- **Check λ_1 :**

$$3 - k < 0$$

$$3 < k$$

$$k > 3$$

Conclusion

For the system to be stable, both conditions ($k > -4$ AND $k > 3$) must be true. The condition $k > 3$ is the stricter one.

Answer:

Yes, the control law stabilizes the system, **provided that $k > 3$** .

Based on the logic from the slides you shared earlier (specifically Slides 30-32 regarding "Assigning Eigenvalues" and Degrees of Freedom), here is the step-by-step solution.

Step 1: Check for Controllability

First, we need to see how many "holes" (states) we can actually reach with the new B matrix.

We calculate the **Controllability Matrix** $\mathcal{C}$.

- **A Matrix:** (from previous exercise)

$$A = \begin{bmatrix} 3 & -1 & 0 \\ 0 & -4 & 2 \\ 0 & 0 & -1 \end{bmatrix}$$

- **B Matrix:**

$$B = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

Calculate the columns:

$$\begin{aligned} 1. \quad B &= \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \\ 2. \quad AB &= \begin{bmatrix} 3-1 \\ -4 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \\ 0 \end{bmatrix} \\ 3. \quad A^2B &= \begin{bmatrix} 6+4 \\ 16 \\ 0 \end{bmatrix} = \begin{bmatrix} 10 \\ 16 \\ 0 \end{bmatrix} \end{aligned}$$

Here is the step-by-step breakdown of the matrix calculation and the logical reasoning for the next steps.

Part 1: How to Calculate $\mathcal{C}$

The Controllability Matrix $\mathcal{C}$ is defined as:

$$\mathcal{C} = [B \quad AB \quad A^2B \quad \dots]$$

Since your system is 3×3 , we need the first three columns: B , AB , and A^2B .

Given:

$$A = \begin{bmatrix} 3 & -1 & 0 \\ 0 & -4 & 2 \\ 0 & 0 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

Step 1: The First Column (B)

This is simply the B matrix itself.

$$\text{Col 1} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

Step 2: The Second Column (AB)

Multiply matrix A by vector B .

$$AB = \begin{bmatrix} 3 & -1 & 0 \\ 0 & -4 & 2 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

- **Row 1:** $(3 \cdot 1) + (-1 \cdot 1) + (0 \cdot 0) = 3 - 1 = \mathbf{2}$
- **Row 2:** $(0 \cdot 1) + (-4 \cdot 1) + (2 \cdot 0) = 0 - 4 = \mathbf{-4}$
- **Row 3:** $(0 \cdot 1) + (0 \cdot 1) + (-1 \cdot 0) = \mathbf{0}$

$$\text{Col 2} = \begin{bmatrix} 2 \\ -4 \\ 0 \end{bmatrix}$$

Step 3: The Third Column (A^2B)

Multiply matrix A by the result we just got (vector AB).

$$A(AB) = \begin{bmatrix} 3 & -1 & 0 \\ 0 & -4 & 2 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ -4 \\ 0 \end{bmatrix}$$

- **Row 1:** $(3 \cdot 2) + (-1 \cdot -4) + (0 \cdot 0) = 6 + 4 = \mathbf{10}$
- **Row 2:** $(0 \cdot 2) + (-4 \cdot -4) + (2 \cdot 0) = 0 + 16 = \mathbf{16}$
- **Row 3:** $(0 \cdot 2) + (0 \cdot -4) + (-1 \cdot 0) = \mathbf{0}$

$$\text{Col 3} = \begin{bmatrix} 10 \\ 16 \\ 0 \end{bmatrix}$$

Step 4: Assemble the Matrix

Put the three columns side-by-side:

$$\mathcal{C} = \begin{bmatrix} 1 & 2 & 10 \\ 1 & -4 & 16 \\ 0 & 0 & 0 \end{bmatrix}$$

Rank Check:

$$\mathcal{C} = \begin{bmatrix} 1 & 2 & 10 \\ 1 & -4 & 16 \\ 0 & 0 & 0 \end{bmatrix}$$

The last row is all zeros, so the Rank is **2**.

Since the system has 3 states ($n = 3$), but the rank is 2, the system is **Not Fully Controllable**.

The Uncontrollable Part:

Looking at the third row of the system equations ($\dot{x}_3 = -x_3$), we see it has no input u . The eigenvalue $\lambda = -1$ is uncontrollable.

- Is it stable? **Yes** (-1 is negative).
- Since it is stable, we don't need to fix it. We only need to assign the eigenvalues for the remaining **2 controllable states**.

This sentence is telling you three separate facts about the third variable (x_3) based on the matrices you are working with.

Here is the breakdown of exactly where each fact comes from:

1. Why is it "Decoupled"?

Look at the **bottom row** of your A matrix.

$$A = \begin{bmatrix} 3 & -1 & 0 \\ 0 & -4 & 2 \\ \mathbf{0} & \mathbf{0} & -1 \end{bmatrix}$$

If you write this row as an equation:

$$\dot{x}_3 = (0 \cdot x_1) + (0 \cdot x_2) + (-1 \cdot x_3)$$

$$\dot{x}_3 = -x_3$$

The Logic:

Notice that x_1 and x_2 are multiplied by **0**.

This means x_3 **does not care** what the rest of the system is doing. It lives in its own bubble. It is physically disconnected (decoupled) from the motion of x_1 and x_2 .

2. Why is it "Uncontrollable"?

Now look at the **bottom row** of your B matrix.

$$B = \begin{bmatrix} 1 \\ 1 \\ \mathbf{0} \end{bmatrix}$$

If we add the input to the equation:

$$\dot{x}_3 = -x_3 + (\mathbf{0} \cdot u)$$

The Logic:

- **Direct Control:** The B matrix has a **0**. This means your input u cannot touch x_3 directly.
- **Indirect Control:** Usually, if you can't touch a variable directly, you try to push a neighbor (x_2) and hope the neighbor pushes x_3 . But because x_3 is **decoupled** (see point #1), it doesn't listen to neighbors either.

Since you can't push it, and neighbors can't push it, it is **Uncontrollable**.

3. Why is the Eigenvalue -1 ?

Because the matrix A is **Triangular** (all numbers below the diagonal are zero), the eigenvalues are simply the numbers sitting on the diagonal.

- Top Left: 3
- Middle: -4
- Bottom Right: -1

The "So What?":

Since the eigenvalue is -1 (a negative number), this uncontrollable state is **Stable**.

Even though you can't control x_3 , it will naturally calm down and go to zero on its own (e^{-1t} decay). This is why we don't need to worry about it and can focus our controller design on just x_1 and x_2 .

Step 2: Determine the Order of the Problem (d_W)

We are now trying to control a system with **2 controllable states**.

This means our "Effective Plant" $P(s)$ is a **2nd Order System**.

- Equation complexity without controller: **2**.

I didn't "assume" it; the math **told us** to ignore the 3rd dimension.

Here is exactly how we concluded that the "Effective Plant" is only **Dimension 2**, even though the matrices are 3×3 .

1. The Rank Test (The Evidence)

In the previous step, we calculated the Controllability Matrix $\mathcal{C}$ and checked its rank.

- **Total States (n):** 3 (Because matrix A is 3×3).
- **Rank of $\mathcal{C}$:** 2 (Calculated in the previous step).

The Rule: The Rank tells you the **Dimension of the Controllable Subspace**.

Even though the robot physically has 3 parts (x_1, x_2, x_3), the input u is mathematically

disconnected from one of them.

2. The "Discarding" Logic

When designing a controller for "Eigenvalue Assignment," you are only responsible for the states you can actually touch.

- **State 1 & 2:** Controllable. We **must** design a controller to fix them. (**Dimension = 2**)
- **State 3:** Uncontrollable. We **cannot** fix it. We just have to hope it is stable (which it is, $\lambda = -1$). Since we can't change it, we remove it from our "To-Do List."

3. The Analogy: The Car with a Loose Coin

Imagine you are driving a car.

- x_1 : Car Position.
- x_2 : Car Velocity.
- x_3 : A coin rattling in the cup holder.

The system has **3 dimensions**.

However, your steering wheel (u) only affects the Car Position and Velocity (x_1, x_2). No matter how you drive, you cannot stop the coin from rattling (x_3 is uncontrollable).

The Problem: "Design a driver to park the car."

The Solution: You only calculate the math for the **2 dimensions** of the car. You ignore the coin because you can't control it anyway.

Summary

We treated the plant as **Dimension 2** because the **Rank was 2**.

If the Rank had been 3, we would have treated it as Dimension 3 (and we would have needed a bigger controller).

Step 3: Test Controller Dimensions (Using $n_{DOF} \geq d_W$)

We need to find the smallest controller dimension (k) where the number of "Knobs" (n_{DOF}) is

equal to or greater than the number of "Roots" (d_W).

Attempt 1: Dimension 0 (Static Controller)

- **Controller:** $G(s) = K$ (Just a number).
- **Knobs (n_{DOF}):** 1 knob (K).
- **System Complexity (d_W):** Plant Order (2) = **2 roots**.
- **Check:** Is $1 \geq 2$? **No**.
- *Result:* A simple gain cannot assign 2 poles arbitrarily.

Attempt 2: Dimension 1 (Dynamic Controller)

- **Controller:** This introduces a structure like $G(s) = \frac{as+b}{s+c}$.
- **Knobs (n_{DOF}):** We have 3 variables to tune (a, b, c).
- **System Complexity (d_W):**
When you multiply a 2nd order Plant by a 1st order Controller, the total system becomes **3rd Order**.

$$d_W = \text{Plant}(2) + \text{Controller}(1) = \mathbf{3 \text{ roots}}$$

- **Check:** Is $3 \geq 3$? **Yes!**

It is not random! That equation is a very specific "standard shape" in mathematics.

To answer your question directly: **No, you cannot just sprinkle s 's wherever you want.**

There are strict rules about where s can go. Here is the logic behind why I wrote it exactly like $\frac{as+b}{s+c}$.

1. The Recipe for "Dimension 1"

In the previous step, we proved we needed a **Dimension 1** controller.

In Control Theory, "Dimension 1" means **the highest power of s must be 1**.

- **Dimension 0 (Static):** K (No s allowed).
- **Dimension 1:** You are allowed to use s^1 .
- **Dimension 2:** You are allowed to use s^2 .

Since we needed Dimension 1, I wrote the **General Form** for a ratio of two 1st-degree polynomials:

$$G(s) = \frac{\text{Polynomial of degree 1}}{\text{Polynomial of degree 1}} = \frac{a \cdot s^1 + b}{1 \cdot s^1 + c}$$

If I had randomly added another s (like s^2), it would have become a **Dimension 2** controller, which is more complex than we needed.

2. Can you take "many s "? (Increasing the Order)

You asked: *"Is that mean I can take many s ?"*

The answer is **Yes, you CAN, but it costs you.**

If you decide to use a **Dimension 2** controller, the equation would look like this:

$$G(s) = \frac{as^2 + bs + c}{s^2 + ds + e}$$

- **The Good News:** You now have **5 knobs** (a, b, c, d, e). You have massive freedom to solve very hard problems.
- **The Bad News:** You have increased the complexity (d_W) of the system.
 - Every time you add an s to the denominator, you are adding a **new energy storage element** (like a capacitor or spring) to your physical circuit.
 - You are making the controller more expensive and harder to build.

The Engineering Rule: Always use the **lowest number of s 's possible** to solve the problem. Don't buy a bus to carry a pizza.

3. Why is there an s on the Top and Bottom?

Why not just $\frac{1}{s+c}$? Or just $as + b$?

- **The Bottom s (The Pole):**

We *must* have an s on the bottom for the system to be physically buildable (this is called being "Proper"). In the real world, systems usually can't react instantly; they need a denominator to represent the time lag.

- **The Top s (The Zero):**

We put an s on top ($as + b$) to give us an extra **Knob**.

- If we used $\frac{1}{s+c}$, we only have **1 knob** (c).
- If we use $\frac{as+b}{s+c}$, we have **3 knobs** (a, b, c).

Summary

I did not place s randomly. I followed the strict formula for a **1st Order Transfer Function**:

$$\frac{\text{Output}}{\text{Input}} = \frac{\text{Zero part } (as + b)}{\text{Pole part } (s + c)}$$

- **s on top** gives you control over the transient speed (derivative action).
- **s on bottom** makes the system stable and buildable (integral/filter action).

Answer

The minimal dimension of the controller $G(s)$ is **1**.

A first-order controller (dimension 1) provides enough degrees of freedom (3 knobs) to place all the poles of the combined closed-loop system (3 roots).